

NOTE 04 — TCP/IP MODEL VS OSI MODEL

What is a network model?

A network model provides a structured way of understanding how devices communicate.

Two important models are:

- OSI Model
- TCP/IP Model

TryHackMe's introductory networking material explains the seven-layer OSI model and contrasts it with the more compact TCP/IP model used in practical networking. TryHackMe

OSI Model

The OSI model contains 7 layers.

7. Application
6. Presentation
5. Session
4. Transport
3. Network
2. Data Link
1. Physical

Layer 7 — Application

Provides network services to applications.

Examples:

- HTTP
- HTTPS
- DNS
- FTP
- SMTP

Layer 6 — Presentation

Responsible for data representation, encoding, encryption, and compression concepts.

Layer 5 — Session

Manages communication sessions between applications.

Layer 4 — Transport

Provides end-to-end communication.

Protocols:

- TCP
- UDP

Layer 3 — Network

Responsible for logical addressing and routing.

Main protocol:

- IP

Layer 2 — Data Link

Handles local network communication and MAC addressing.

Examples:

- Ethernet
- Wi-Fi

Layer 1 — Physical

Deals with physical transmission.

Examples:

- Cables
- Radio signals
- Electrical signals
- Network hardware

TCP/IP Model

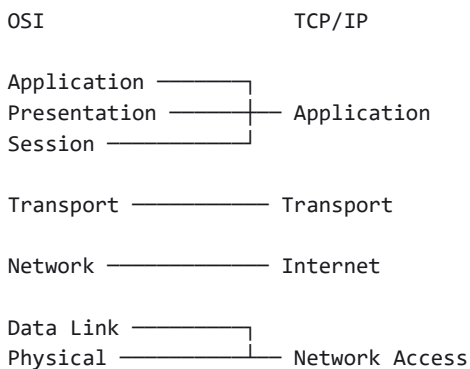
The TCP/IP model is commonly represented using four layers.

TCP/IP Layer	Main responsibility	Example
Application	Application/network services	HTTP, DNS, SMTP
Transport	End-to-end communication	TCP, UDP
Internet	IP addressing/routing	IPv4, IPv6
Network Access	Local/physical communication	Ethernet, Wi-Fi

OSI vs TCP/IP

OSI	TCP/IP
7 layers	4 layers
Mainly conceptual/reference model	Practical networking model
More detailed	More compact
Developed by ISO	Developed around DoD/Internet protocols
Useful for learning/troubleshooting	Widely used in real networking

Layer mapping

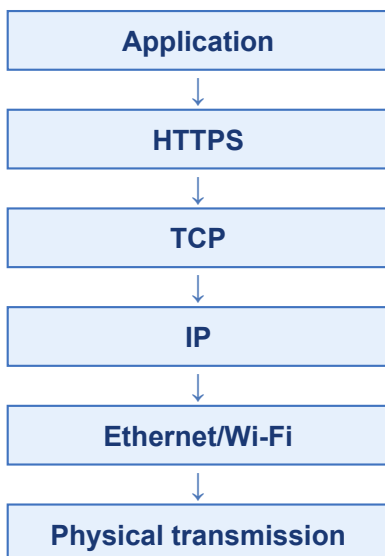


Example: Opening a Website

Suppose you visit:

<https://example.com>

A simplified process is:



The receiving system processes the information in the reverse direction.

TCP vs UDP

TCP	UDP
Connection-oriented	Connectionless
Reliable delivery	No built-in delivery guarantee
Uses acknowledgements	No acknowledgements in the same sense
More overhead	Lower overhead
Example: HTTPS traffic	Example: DNS/streaming use cases